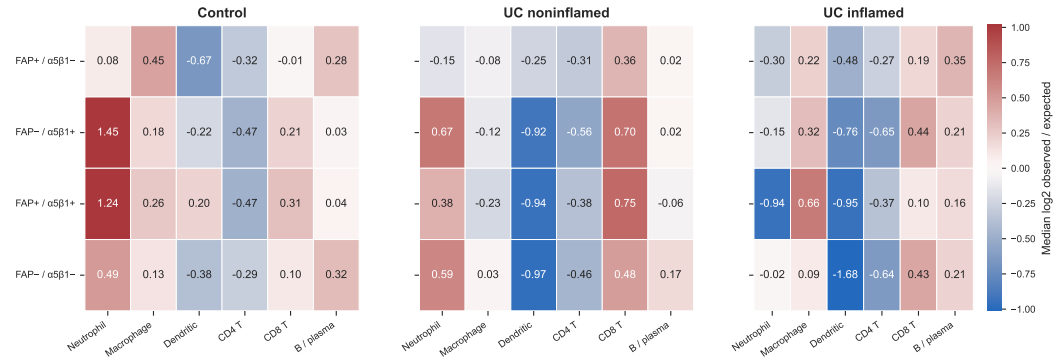


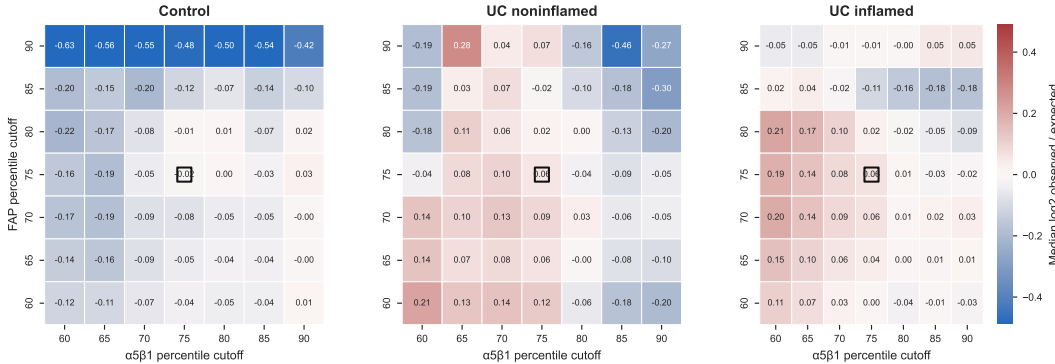
a Immune-lineage spatial specificity

Median log2 observed/expected proximity across FAP/ $\alpha 5\beta 1$ strata



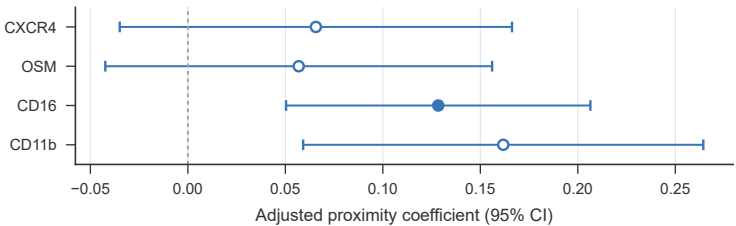
b Threshold sensitivity

FAP and $\alpha 5\beta 1$ percentile cutoffs across tissue conditions



c Adjusted neutrophil proximity effects

Filled point: restricted spatial-null $q < 0.05$; bars show 95% CI



Model q	Spatial-null q
0.242	0.844
0.242	0.843
0.008	0.032
0.008	0.144